

Yin-Ching (William) Lee

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Research Interests

Reinforcement learning, Bayesian inference, sequential decision-making under uncertainty, robotics, reasoning.

Education

Boston University	Boston, MA
<i>Master of Science in Computer Science</i>	Sep 2024 – May 2025
Boston University	Boston, MA
<i>Bachelor of Arts in Computer Science</i>	Sep 2021 – May 2024

Research Experience

Machine Learning Research Assistant	Boston, MA
<i>Pacchiano's Lab for Adaptive and Intelligent Algorithms (PLAIA), Boston University</i>	Aug 2025 – Present
Advisors: Aldo Pacchiano, Alessio Russo	

- Developed C-ICPE, a theory-guided learned model for active sequential testing (pure exploration) that identifies an unknown ground-truth hypothesis in a continuous recommendation space using as few adaptive queries as possible
- Meta-trained sequential neural architectures over a task prior to jointly learn exploration, stopping, and recommendation strategies, enabling the model to actively gather evidence at inference time and output an ϵ -optimal recommendation without further parameter updates
- Designed and validated the GeoChemBandit benchmark using USGS geochemical survey data (copper concentrations) with a Gaussian Process + FFT circulant embedding formulation for 2D spatial active learning
- Benchmarked C-ICPE against TPE, CMA-ES, GP-UCB, and GP-logEI baselines on ϵ -best-arm identification, and Ackley function tasks, using hierarchical bootstrap (BCa) confidence intervals for statistical evaluation

Machine Learning Research Assistant	Bologna, Italy
<i>Extreme Computing System (ECS) Lab, University of Bologna</i>	June 2024 – Aug 2025
Advisors: Andrea Bartolini, Andrea Acquaviva	

- Investigated autonomous agile flight in dynamic environments using Spiking Neural Networks (SNNs)
- Integrated SNNs and a quadrotor environment into the Isaac Gym simulator, achieving a 2.47% improvement in success rate and a 28.57% reduction in completion time for navigating through five swinging gates, compared to traditional ANNs

Machine Learning Research Assistant	Taipei, Taiwan
<i>Reinforcement Learning and Games Lab, Academia Sinica</i>	Oct 2023 – May 2024
Advisor: Ti-Rong Wu	

- Tackled planning problems using Deep Reinforcement Learning and the AlphaZero algorithm
- Devised a novel internal/external curriculum learning algorithm, raising AlphaZero's success rate from 25% to 56%
- Incorporated a Proof-Cost Network for difficulty prediction on Sokoban instances to optimize the curriculum learning process

Machine Learning Research Assistant, HPC Systems Performance Analytics	Boston, MA
<i>Performance and Energy Aware Computing (PEAC) Lab, Boston University</i>	Feb 2023 – Sep 2024
Advisor: Ayse Coskun	

- Focused on machine learning applications for HPC systems performance analytics; funded by Sandia National Laboratories

- Engineered a robust unsupervised anomaly-detection framework, achieving an 88% F1-score despite 10% training data contamination
- Implemented a state-of-the-art GAN + VAE model from scratch and conducted a comparative analysis against baseline approaches

Publications

*Equal contribution indicated with *.*

- Russo, Alessio*, Yin-Ching Lee*, Ryan Welch, and Aldo Pacchiano. "In-Context Learning for Pure Exploration in Continuous Spaces." *arXiv*, 2026, arXiv:2602.17976. Presented at two ICML 2026 workshops (one with oral presentation); extended version under review, NeurIPS 2026.
- Lee, Yin-Ching, Sebastiano Mengozzi, Luca Zanatta, Andrea Bartolini, Andrea Acquaviva, and Francesco Barchi. "Bio-Inspired Drone Control: A Reinforcement Learning-Trained Spiking Neural Networks for Agile Navigation in Dynamic Environment." *Proceedings of the 2025 IEEE International Conference on Omni-Layer Intelligent Systems (COINS)*, IEEE, 2025, pp. 1–8.
- Sencan, Efe, Yin-Ching Lee, Connor Casey, Benjamin Schwaller, Vitus J. Leung, Jim Brandt, Brian Kulis, Manuel Egele, and Ayse K. Coskun. "Refine: A Robust Approach to Unsupervised Anomaly Detection for Production HPC Systems." *Proceedings of ISC High Performance (ISC-HPC)*, 2025.
- Sencan, Efe, Yin-Ching Lee, Connor Casey, Benjamin Schwaller, Vitus J. Leung, Jim Brandt, Brian Kulis, Manuel Egele, and Ayse K. Coskun. "Refine: Robust Unsupervised Anomaly Detection for Production HPC Systems." *Deployable AI (DAI) Workshop, AAAI 2025*.

Posters & Presentations

- Project poster, *ISC High Performance (ISC-HPC)*, 2024
- Research poster, *Design, Automation & Test in Europe (DATE)*, 2023
- Tutorial presentation, *LDMS Conference (LDMSCON)*, 2024

Other Professional Experience

Technical Project Manager

BU Spark!

Boston, MA

Jan 2024 – May 2024

- Oversaw technical aspects of cross-institutional projects with MIT Lincoln Lab and Harvard Herbaria; conducted code reviews to ensure client-ready deliverables

Machine Learning Engineer Intern

BU Spark! & Atfal Mafkoda

Boston, MA

Jan 2023 – May 2023

- Built a face-detection and aging pipeline to help locate missing children in Egypt, achieving 76.8% face-detection accuracy on low-quality images; received the Audience Choice Award at the BU Experiential Learning Expo

Software Engineer Intern

Yi-Yang Construction Co., Ltd.

Changhua, Taiwan

May 2022 – Sep 2022

- Developed a Swift-based clock-in/leave-request app with facial recognition, serving 80+ users; automated monthly reporting, reducing weekly workload by 10+ hours

Technical Skills

Languages & Frameworks: Python, Java, Dart, JavaScript, C++, Swift, SQL, PyTorch, TensorFlow, Scikit-learn, MATLAB

Systems & Tools: ROS, SGE/UGE job scheduling, AWS, GCP, Firebase, Isaac Gym